

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы



Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.



- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы



```
irabzalimov@irabzalimov:~$ ls /etc/ > file.txt
irabzalimov@irabzalimov:~$ ls >> file.txt
irabzalimov@irabzalimov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
```

Рисунок 1: Запись в файл



```
irabzalimov@irabzalimov:~$ grep .conf file.txt > conf.txt
irabzalimov@irabzalimov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
```

Рисунок 2: Поиск расширения .conf



```
al/stage04/report/bib/cite.bib
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage04/presentation/_resources/css
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage05/report/_resources/csl
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage05/report/bib/cite.bib
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage05/presentation/_resources/css
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage06/report/_resources/csl
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage06/report/bib/cite.bib
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-person
al/stage06/presentation/_resources/css
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/r
eport/_resources/csl
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/r
eport/bib/cite.bib
/home/irabzalimov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/p
resentation/_resources/css
/home/irabzalimov/git-extended/.git/hooks/commit-msg.sample
/home/irabzalimov/git-extended/.git/config
/home/irabzalimov/conf.txt
irabzalimov@irabzalimov:~$
```

Рисунок 3: Поиск файлов



```
find: '/etc/cvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: '/etc/openvpn/server': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: '/etc/userdb': Permission denied
```

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Рисунок 4: Поиск файлов



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ find ~ -name "log*" > logfile &  
[1] 6974  
irabzalimov@irabzalimov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
irabzalimov@irabzalimov:~$ rm logfile  
irabzalimov@irabzalimov:~$
```

Рисунок 5: Фоновый запуск процесса



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ gedit &  
[1] 7023  
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ ps | grep gedit  
    7023 pts/1    00:00:00 gedit  
irabzalimov@irabzalimov:~$ kill 7023  
[1]+  Terminated                  gedit  
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса



```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df



```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df



```
irabzalimov@irabzalimov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 26198228 120838300  18% /
devtmpfs         4010528        0   4010528   0% /dev
tmpfs            4034080        92   4033988   1% /dev/shm
tmpfs            1613636       1948   1611688   1% /run
tmpfs             1024          0        1024   0% /run/credentials/systemd-journald.service
tmpfs            4034084       128   4033956   1% /tmp
/dev/nvme0n1p3 155186176 26198228 120838300  18% /home
/dev/loop1       115712    115712        0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2        49280    49280        0 100% /var/lib/snapd/snap/snapd/25935
/dev/nvme0n1p2  1992552   516696   1354616   28% /boot
/dev/loop0        68480    68480        0 100% /var/lib/snapd/snap/core24/1349
tmpfs             1024          0        1024   0% /run/credentials/systemd-resolved.service
tmpfs            806816       160    806656   1% /run/user/1099
irabzalimov@irabzalimov:~$
```

Рисунок 9: Справка по команде du



```
4      ./git-extended/.git/objects/d5
4      ./git-extended/.git/objects/61
4      ./git-extended/.git/objects/2e
4      ./git-extended/.git/objects/51
4      ./git-extended/.git/objects/5b
4      ./git-extended/.git/objects/84
4      ./git-extended/.git/objects/1b
72     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
208    ./git-extended/.git
216    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
1031884 .
irabzalimov@irabzalimov:~$
```

Рисунок 10: Запуск команды du



```
/home/irabzalimov/git-extended/.git/objects/d4
/home/irabzalimov/git-extended/.git/objects/78
/home/irabzalimov/git-extended/.git/objects/d5
/home/irabzalimov/git-extended/.git/objects/61
/home/irabzalimov/git-extended/.git/objects/2e
/home/irabzalimov/git-extended/.git/objects/51
/home/irabzalimov/git-extended/.git/objects/5b
/home/irabzalimov/git-extended/.git/objects/84
/home/irabzalimov/git-extended/.git/objects/1b
/home/irabzalimov/git-extended/.git/logs
/home/irabzalimov/git-extended/.git/logs/refs
/home/irabzalimov/git-extended/.git/logs/refs/heads
/home/irabzalimov/git-extended/.git/logs/refs/remotes
/home/irabzalimov/git-extended/.git/logs/refs/remotes/origin
/home/irabzalimov/monthly
/home/irabzalimov/reports
/home/irabzalimov/reports/monthly
/home/irabzalimov/reports/monthly/monthly
/home/irabzalimov/ski.plases
/home/irabzalimov/ski.plases/equipment
/home/irabzalimov/ski.plases/plans
/home/irabzalimov/australia
/home/irabzalimov/play
/home/irabzalimov/play/games
/home/irabzalimov/play/games/play
irabzalimov@irabzalimov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе



В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.